10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 8 — Chords and Arcs of a Circle

Unit 8 | Review + MCQs | New Book (PECTAA)

Review Exercise 8 (Solved)

The review covers the whole unit: parts of a circle, the perpendicular from the centre to a chord, equal chords and equal arcs, and the circle through three non-collinear points.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) Distance of a point on the circumference to the centre of the circle is called:

(a) radius (b) arc
(c) chord (d) tangent

Answer: (a) radius — every point of the circle is at the same distance from the centre

(ii) Radii of the same circle are:

(a) all unequal (b) all equal
(c) half of each chord (d) double of the diameter

Answer: (b) all equal — a circle has one fixed radius

(iii) The boundary of the circle is called:

(a) chord (b) segment
(c) circumference (d) diameter

Answer: (c) circumference — the curve that goes round the circle is the circumference

(iv) Any part of a circumference is called:

(a) chord (b) diameter
(c) radius (d) arc

Answer: (d) arc — arc is the piece of the circumference between two points

(v) A chord passing through the centre of the circle is called:

(a) radius (b) diameter
(c) secant (d) circumference

Answer: (b) diameter — the longest chord of a circle is the diameter

(vi) In the given figure A, B, C lie on the circle and O is the centre, what is AB?

(a) diameter (b) tangent
(c) chord (d) arc

Answer: (c) chord — AB joins two points of the circle and does not pass through O

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(vii) In the same figure, the major arc is:

(a) arc AB (b) arc BC
(c) arc BDC (d) arc AD

Answer: (c) arc BDC — the major arc is the longer arc, taken through the third point

(viii) How many circle(s) can pass through three non-collinear points?

(a) one (b) two
(c) three (d) many

Answer: (a) one — three non-collinear points fix one and only one circle

(ix) Perpendicular bisector of a chord always passes through the _____ of the circle.

(a) arc (b) radius
(c) centre (d) circumference

Answer: (c) centre — the perpendicular from the centre bisects every chord

(x) The sum of the measures of the central angles of a circle is:

(a) 90° (b) 180°
(c) 270° (d) 360°

Answer: (d) 360° — one complete turn round the centre is 360°

2. On a circular clock with a radius of 6 cm, the points from 2 to 10 form a chord that is 10 cm long. Find the perpendicular distance from the centre of the clock to the chord.

Chord = 10 cm, r = 6 cm, half chord = 5 cm

$d^2 = r^2 - (\text{half chord})^2 = 6^2 - 5^2 = 36 - 25 = 11$

$d = \sqrt{11} = 3.32$ cm

Answer: Perpendicular distance ≈ 3.32 cm

3. A chord 6 cm long is at a distance of 4 cm from the centre. Calculate the radius of the circle.

Half chord = 3 cm, d = 4 cm

$r^2 = d^2 + (\text{half chord})^2 = 4^2 + 3^2 = 16 + 9 = 25$

r = 5 cm

Answer: Radius of the circle = 5 cm

4. In a circular park, two benches are placed so that the chords formed by their positions are both 10 m long. What can you say about the angles subtended at the centre by each bench?

Both chords are 10 m long, so they are equal chords

Equal chords of a circle subtend equal angles at the centre

Answer: The two angles at the centre are equal

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5. Jannat is designing a triangular garden with corners at points A(2, 1), B(4, 4) and C(1, 5). Can she install a circular fountain that touches all three corners?

Slope of AB = $\frac{4-1}{4-2} = \frac{3}{2}$

Slope of BC = $\frac{5-4}{1-4} = -\frac{1}{3}$

Slope of AB ≠ slope of BC

So A, B and C are not collinear

Through three non-collinear points one and only one circle passes

Answer: Yes, one circular fountain can touch all three corners

6. Two chords, PQ and RS, each measure 18 cm in length. If the distance of PQ from the centre of the circle is 7 cm, find the distance of RS from the centre.

PQ = RS = 18 cm, so the chords are equal

Equal chords are equidistant from the centre

Answer: Distance of RS from the centre = 7 cm

7. A tree branch lies across a circular pond, forming a 20 m chord. A measuring rod of length 10 m stands perpendicular to the chord from the centre. What is the radius of the pond?

Half chord = $\frac{20}{2}$ = 10 m, d = 10 m

$r^2 = d^2 + (\text{half chord})^2 = 10^2 + 10^2 = 200$

$r = \sqrt{200} = 14.14$ m

Answer: Radius of the pond ≈ 14.14 m

8. A steel bar 6 m long lies inside a circular structure with ends on the circle. It is bisected by a perpendicular rod from the centre. Find the radius of the circular structure if the perpendicular distance from the centre to the bar is 4 m.

Half chord = $\frac{6}{2}$ = 3 m, d = 4 m

$r^2 = 4^2 + 3^2 = 16 + 9 = 25$

r = 5 m

Answer: Radius of the structure = 5 m

Quick Revision — Unit 8 at a Glance

Radius, diameter, chord, arc, circumference, segment, sector, semi-circle.

Perpendicular from the centre bisects the chord: $r^2 = d^2 + (\text{half chord})^2$.

Equal chords ↔ equal minor arcs ↔ equal central angles ↔ equally far from the centre.

Central angle = measure of its arc; the four central angles of a circle add to 360°.

One and only one circle passes through three non-collinear points.